

केन्द्रीय विद्यालय संगठन, कोलकाता संभाग
KENDRIYA VIDYALAYA SANGATHAN, KOLKATA REGION
प्री-बोर्ड परीक्षा / PRE-BOARD EXAMINATION – 2024-25

कक्षा / CLASS – X
विषय/SUB. – SCIENCE(086)

अधिकतम अंक /MAX. MARKS-80
समय/TIME – 03 घंटे/Hours

MARKING SCHEME

Section-A		
1.	C. $2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$	1
2.	D. by a burning candle	1
3.	D. carbon dioxide in human cells	1
4.	C. Posture and balance	1
5.	B. $2\text{HCl} + \text{Ba}(\text{OH})_2 \rightarrow \text{BaCl}_2 + 2\text{HOH}$	1
6.	D. Ureter	1
7.	B. Cooking of food.	1
8.	A. All variations in a species have equal chance of survival	1
9.	C. (ii) only Alternate question for VI A. Slaked lime	1
10.	A. Tallness is the dominant trait	1
11.	D. Magnesium and calcium	1
12.	C Alternate question for VI B. The current will change its direction continuously	1
13.	C. Alkanes	1
14.	B. Will increase	1

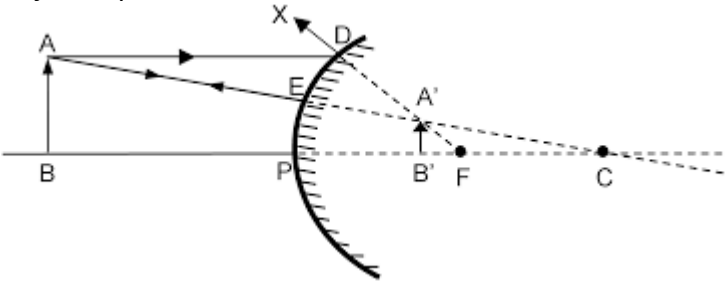
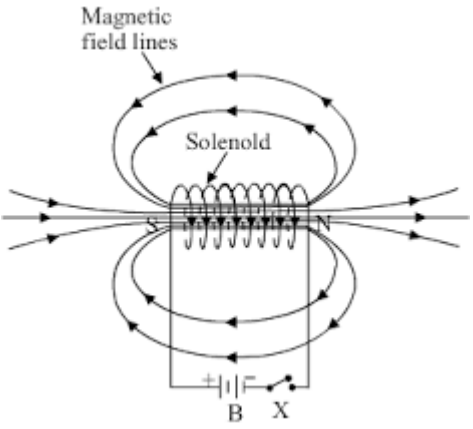
15.	B. Ammeter should be connected in series and voltmeter in parallel	1
16.	C. Plumule, cotyledon and radicle Alternate question for VI B. multiple fission	1

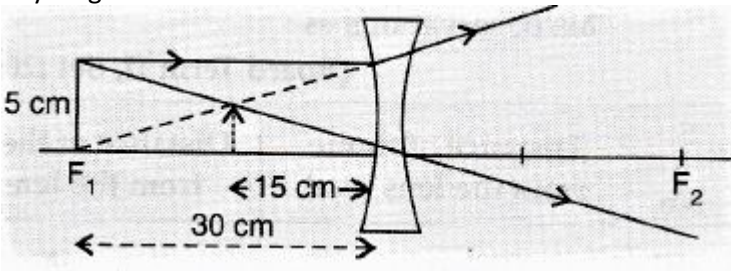
17.	C	1
18.	B	1
19.	A	1
20.	D	1

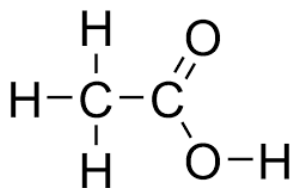
Section-B

21.	(a) Ferric oxide/ Fe_2O_3 and Al (b) thermite reaction. $\text{Fe}_2\text{O}_3 (\text{s}) + \text{Al} (\text{s}) \rightarrow \text{Al}_2\text{O}_3 + 2 \text{Fe}(\text{l}) + \text{Heat}$	1 1
22.	(a)Respiration. The process of acquiring oxygen from outside the body, and to use it in the process of break-down of food sources for cellular needs, is what we call respiration. (b) Lactic acid is a byproduct of anaerobic respiration, which is when the body produces energy without oxygen. A buildup of lactic acid can cause muscle cramps.	1 1
23.	Y-MORE WAVELENGTH X-LESS WAVELENGTH Dispersion is defined as the spitting of white light into different colours when passed through a prism. Alternate question for VI Splitting of white light into seven colours-dispersion • Conditions- The sun should be behind • There should be rain, fog, or some other source of water droplets in front(any one)	0.5 0.5 1 1 1
24.	(a) Peristaltic movement is the contraction and relaxation of these muscles in the food pipe which brings the food down the pipe and into the stomach. (b) They store the CO_2 in their cells until the sun comes out and they can carry on with photosynthesis during the day time. OR, A mechanism in which blood circulates twice through the heart in one complete cycle is known as double circulation. Double circulation is	1 1 1

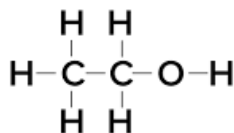
	necessary in human beings to separate oxygenated and deoxygenated blood as this enhances the efficiency of cellular respiration in humans and consequently helps in maintaining a constant body temperature.	1
25.	(a) The food chain including the above mentioned organisms will be: Cereal plant → Human being (b) Possible disadvantage of cereal growing in pesticide rich soil is that it will accumulate chemicals in the successive trophic levels.	1 1
26.	If the plant part moves in the direction of gravity, it is called positive geotropism. If the plant part moves against the direction of gravity, it is negative geotropism. Alternate question for VI Reflex arc is evolved in animals because the thinking process of the brain is not fast enough. Reflex arc enables the body to give quick responses to harmful stimuli so that chances of damage to body are decreased.	1 1 2
Section-C		
27.	Procedure: 1. Take magnesium ribbon with a pair of tongs and burn it in flame in the presence of air. 2. Collect the product formed and dissolve it in warm water. 3. Add red litmus paper into it. 4. Litmus changes its colour to blue. 5. Burn sulphur in a deflagrating spoon in the presence of air and dissolve the oxide formed in water. 6. Dip blue litmus paper and its colour changes to red.	0.5 0.5 0.5 0.5 0.5 0.5
28.	A. On addition of saliva which contains an amylase, a starch digesting enzyme (chemical compound that catalyses the breakdown of starch); the starch gradually converts to sugar. B. Gall Bladder As the entry of bile is blocked the formation of small fat globules do not occur from the dietary fats. OR, (a) The above experiment is conducted to show that chlorophyll is necessary for photosynthesis. (b) The plant was placed in the dark before beginning the experiment to destarch the leaves. (c) Balanced chemical equation representing the process of photosynthesis is given below:	1.5 1.5 1 1 1

	<u>Equation depicting the reaction taking place during photosynthesis</u> $6\text{CO}_2 + 12\text{H}_2\text{O} \xrightarrow[\text{Sunlight}]{\text{Chlorophyll}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$	
29.	a. sodium b. $2\text{Na (s)} + 2\text{H}_2\text{O (l)} \rightarrow 2\text{NaOH (aq)} + \text{H}_2 \text{ (g)}$ $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ c. Electrolytic reduction: Electric current is passed through molten NaCl. Sodium is formed at cathode and chlorine gas is liberated at anode.	1 1 1
30.	The ratio of the focal lengths of mirrors AB and CD will be equal to one as both the mirrors are part of same spherical ball. Hence, their radius of curvature and focal length will be same. The mirror AB will always form a virtual image when object is placed in front of it because mirror AB is a convex mirror.  Alternate question for VI Four characteristics of the image formed by the given convex mirror are (Any Three): (i) Virtual (ii) Erect (iii) Diminished (iv) Image is always formed behind the mirror between pole and focus.	1 1 1
31.	A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder is called solenoid.  Use-to magnetise a piece of magnetic material like soft iron when placed inside the coil.	1 1 1

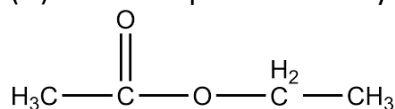
32.	a) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ $\Rightarrow \frac{1}{u} = \frac{1}{v} - \frac{1}{f}$ $\Rightarrow u = \frac{vf}{f-v} = \frac{-15 \text{ cm} \times -30 \text{ cm}}{-30 \text{ cm} - (-15 \text{ cm})}$ b) $= -30 \text{ cm}$ $m = \frac{v}{u} = \frac{h_2}{h_1}$ $h_2 = \frac{v}{u} \times h_1$ $= \frac{-15 \text{ cm}}{-30 \text{ cm}} \times 5 \text{ cm}$ =2.5 cm. Ray diagram- 	0.5 0.5 0.5 0.5 1
33.	(a) i. 10% of x = 100 x = 1000 J in deer 10% of y = 1000 y = 10,000 J in Plant ii. The population of tigers decreases, and the population of grass increases. (b){any one} -The biodegradable wastes emit a foul smell that can be dangerous to people when inhaled because of how slowly they decompose. -The dumping sites may serve as a breeding ground for dangerous organisms that can harm plants, animals and humans. -Depletion of oxygen could occur as aquatic organism populations grow.	1 1 1
SECTION D		
34.	A.(a)Carboxylic acid having molecular formula C ₂ H ₄ O ₂ is (acetic acid or ethanoic acid. Its structure is	1



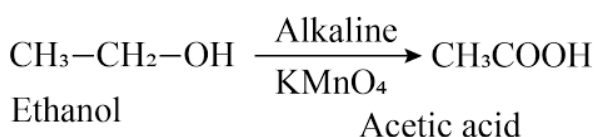
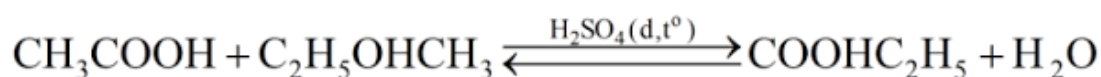
(b) Since, an alcohol which oxidation with alkaline KMnO_4 followed by acidification gives ethanoic acid. Therefore, it must be ethanol. Its structure is



(c) Since compound X is ethyl Ethanoate(Ester). Its structure is



B. Reactions:



OR,

Explain the given reactions with the examples

(a) Hydrogenation reaction- The addition of hydrogen to the unsaturated molecule to make it saturated is known as hydrogenation.(suitable reaction)

(b) Oxidation reaction- The reactions in which an oxidising agent supply nascent oxygen for oxidation are called oxidation reactions.(suitable reaction)

(c) Substitution reaction- When one atom or a group of atoms replaces or substitutes another atoms or a group of atoms from the molecule, it is known as substitution reaction(suitable reaction)

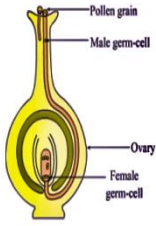
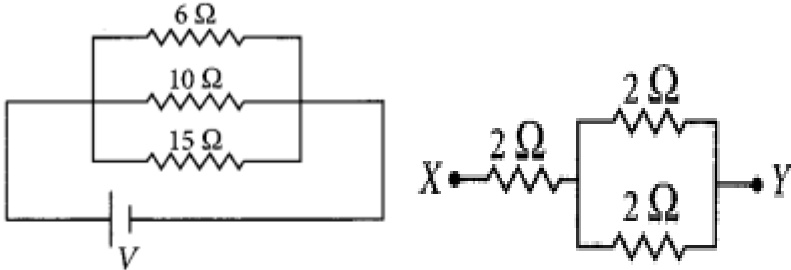
(d) Saponification reaction- When esters are hydrolysed in the presence of a base (NaOH) then the reaction is called saponification reaction.(suitable reaction)

(e) Combustion reaction- Organic compounds burn readily in air to form CO_2 and water vapour along with lot of heat.(suitable reaction)

35.

(i) Nutrients/glucose/oxygen/waste. (any two)

(ii) less surface area for nutrients (glucose/oxygen) to pass from mother to embryo slow growth.

	(iii) uterus; has thick lining with rich supply of blood to nourish the embryo. a) male child b) misused as if the foetus is female, some people engage in aborting the child leading to female foeticide. OR, (a) Pollen grains. Cross-pollination is defined as the deposition of pollen grains from a flower to the stigma of another flower. (b) Pollen tube. They transport the male gamete cells from the stigma to the ovules. (c) C- male gamete D- female gamete (d) E- ovary, F- zygote. (e) 	1 1 1 1 1 1 1
36.	(a) Power is the rate of energy consumption. 1 watt is the power of an appliances which consumes energy at the rate of 1 joule per second. (b) i) Power = $P = VI$ $= 5 \times 0.5$ $= 2.5W$ ii) Resistance (R) = $V/I = 50/5 = 10 \Omega$ iii) energy consumed- $2 \times 2.5 \times 1/1000 = 0.005 KWh$ (b) Find the equivalent resistance of the following circuits.  1st part- $1/R = 1/6 + 1/10 + 1/15 = 10/30 = 1/3$ $R = 3 \text{ ohm.}$ 2nd part- $R_1 = (1/2 + 1/2) \text{ ohm} = 1 \text{ ohm.}$ $R_2 = 2 \text{ ohm}$ Total $R = R_1 + R_2 = 3 \text{ ohm.}$ Alternate question for VI (a) Joule's law about heating effect of electric current: When a current flows through a conductor, the quantity of heat produced in it is directly proportional to (1) the square of the current (2) The resistance of the conductor (3) the time for which the current flows.	2 1 1 1 1 1 1 1

SECTION-E

37.	(a) E, It is displacing most of the metals from their salt solution. (b) Brown deposits of copper as it will be displaced by metal B. (c) $D < C < A < B < E$ (d) D, Its not reacting with salt solution. OR, (d) $Al_2(SO_4)_3$ and $MgSO_4$ as they are not reacting with any metals.	1 1 1 1
38.	(a) (iv) A combination of X and Y chromosomes from either of its parents (b) ii) 50% (c)ii) False OR, (c)1 pair (d) This means that the sex of the child is not determined by the genes inherited from the parents.	1 1 1 1
39.	(a) The refractive index is a measurement of how much a light ray bends when it passes from one medium to another. (b) $n = c/v; = (3 \times 10^8 / 2.5 \times 10^8) \text{ m/s}; = 1.2$ (c) (i) A rectangle glass slab Alternate question for VI The ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant for the pair of the given media. (d) The bending will not be the same as kerosene and oil are denser than water, due to variation in optical density.	1 1 1 1